

RESEARCH ARTICLE

New primers to amplify the fungal ITS2 region – evaluation by 454-sequencing of artificial and natural communities

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Abstract

With recent methodological advances, molecular markers are increasingly used for semi-quantitative analyses of fungal communities. The aim to preserve quantitative relationships between genotypes through PCR places new demands on primers to accurately match target sites and provide short amplicons. The internal transcribed spacer (ITS) region of the ribosome encoding genes is a commonly used marker for many fungal groups. Here, we describe three new primers – fITS7, gITS7 and fITS9, which may be used to amplify the fungal ITS2 region by targeting sites in the 5.8S encoding gene. We evaluated the primers and compared their performance with the commonly used ITS1f primer by 454-sequencing of both artificially assembled templates and field samples. When the entire ITS region was amplified using the ITS1f/ITS4 primer combination, we found strong bias against species with longer amplicons. This problem could be overcome by using the new primers, which produce shorter amplicons and better preserve the quantitative composition of the template. In addition, the new primers yielded more diverse amplicon communities than the ITS1f primer.

Introduction

During the last 20 years, analysis of microbial communities based on amplification and sequencing of genetic markers has revolutionized fungal ecology (Horton & Bruns, 2001; Hibbett *et al.*, 2009). High-throughput methods, such as 454-pyrosequencing (Margulies *et al.*, 2005), enable sequencing of complex fungal communities without prior cloning of amplicons and open up new possibilities to identify community components, even at low relative abundances. The high sequence output may be used to quantify relative abundances of amplicons, assuming that amplicon composition reflects the abundance of specific templates in the samples. Even though translation between genetic markers, biomass and activity is often problematic (Amend *et al.*, 2010), these new sequencing methods open up new avenues for ecological research by providing detailed and at least semi-quantitative information on community composition. A future development towards sequencing of environmental DNA and RNA without prior PCR amplification may be

envisaged (Urich *et al.*, 2008), but current analysis of microbial communities depends on reliable PCR primers to target specific genetic markers.

Molecular identification of fungi largely relies on amplification of the internal transcribed spacer (ITS) regions of the ribosome encoding genes. The ITS regions have high evolutionary rates and are flanked by highly conserved regions with suitable target sites for universal primers (Begerow *et al.*, 2010). Recently, the ITS region was selected as the universal genetic barcode for fungi (Schoch *et al.*, 2012). PCR-based methods were early used to identify pathogenic and mutualistic fungi, growing in association with plant hosts (White *et al.*, 1990; Gardes & Bruns, 1993). Analyses were then restricted to samples strongly dominated by single fungal genotypes, and a major challenge was to avoid amplification of the plant host DNA and co-colonizing fungi, to obtain a single, fungal PCR product. The primers ITS1f (Gardes & Bruns, 1993), which targets a site in the ribosomal small subunit (SSU) encoding region, and ITS4 (White *et al.*, 1990), which targets an ITS-flanking site in the ribosomal large

subunit (LSU) encoding region, were devised based on the scarce sequence information that was available at the time. The ITS1f primer was designed based on nine fungal sequences only, whereof six belonged to Boletales! With this in mind, these primers have proven highly successful in amplifying the ITS region from a wide range of fungal diversity, including representatives of all fungal phyla. These primers have been used extensively, not only in ecological studies, but also to yield phylogenetic markers for taxonomical studies. As databases of reference sequences build up, ITS-sequencing has often turned out to be more reliable than morphological features to identify fungal fruit bodies and cultures.

Whereas the currently used ITS primers were developed to yield PCR products from mono-specific samples, amplification of complex fungal communities adds new challenges. When communities are amplified, there is competition for primers between different templates, and also minor mismatches may disfavour taxa to the extent that they are outcompeted from the amplicon pool. Comparison with sequence databases suggests that the currently used ITS primers are hampered with mismatches relative to their target sequences for many fungal taxa (Bellemain *et al.*, 2010). Sequencing of environmental samples indicates that a significant fraction of fungal diversity remains to be discovered and described (Hibbett *et al.*, 2011), and by relying on primers that were developed based on limited information on primer site variation, there is a risk that important novel branches on the fungal tree of life remain undiscovered. Furthermore, the access to general primers without mismatches becomes pivotal when the composition of a PCR product is to be interpreted quantitatively and amplification bias between species has to be minimized.

The primer ITS1f, which is commonly used in combination with the ITS4 primer to produce amplicons that span the entire ITS region, is highly successful in amplifying fungal sequences even in the presence of large amounts of DNA from other organisms. However, for many fungal species, the ITS1f primer displays mismatches in relation to its target site (Bellemain *et al.*, 2010). Therefore, we present three alternative

primers – fITS7, gITS7 and fITS9, which are based on close to 140 000 fungal ITS sequences. The new primers target binding sites in the 5.8S region, which is situated between the two ITS regions (Fig. 1). This means that, in combination with the ITS4 primer, the new primers yield amplicons that span the ITS2 region only and are considerably shorter than ITS1f-ITS4 fragments, which also include the ITS1 region and the entire 5.8S.

Although the ITS2 region contains less genetic information than the entire ITS, shorter amplicons lead to higher PCR efficiencies, meaning that fewer cycles are needed to obtain the desired product concentration. This is highly advantageous when dealing with problematic samples, where successful amplification within a reasonable number of cycles may be hampered by scarcity of template or high inhibitor concentrations. That product length has to be restricted when PCR is used for quantitative estimation has long been recognized in the context of real-time PCR, where amplicons longer than 200 bp are discouraged (Bio-Rad Laboratories Inc, 2006). For prokaryotic SSU amplicons, it has been demonstrated that increasing length of the target amplicon has a significant negative effect on assessments of microbial richness and also biases community composition (Huber *et al.*, 2009; Engelbrektson *et al.*, 2010). Furthermore, when amplicons span both ITS regions, with the highly conserved 5.8S region in the middle, PCR chimeras are frequently formed where partial fragments derived from different parental templates combine at the conserved 5.8S region and yield sequences with the two ITS regions originating from different templates. Chimeras cause severe problems in subsequent sequence clustering and identification (Nilsson *et al.*, 2010; Quince *et al.*, 2011) but are less likely to occur when amplification is restricted to one of the ITS regions. When attempting to amplify small amounts of fungal DNA from insect samples using the ITS1f/ITS4 primer combination, 19% of the resulting sequences were chimeric with a fungal ITS1 region combined with an ITS2 region of insect origin (Y. Strid, unpublished).

The new primers were tested on an artificially assembled community of PCR products, and the results were

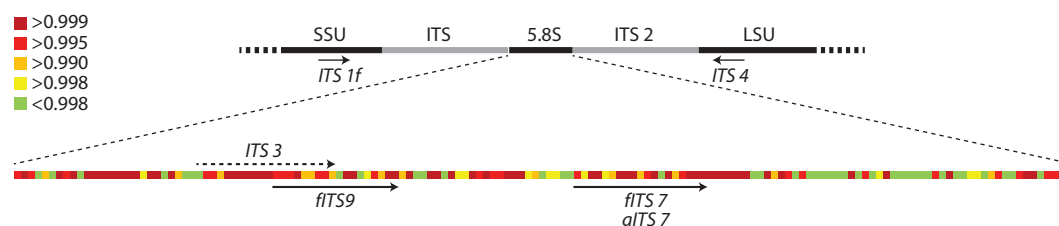


Fig. 1. Organization of ribosomal genes and target regions of PCR primers. Colours in the close-up of the 5.8S region indicate the frequency of the most common nucleotide among close to 140 000 aligned fungal sequences.

evaluated by 454-sequencing. Primer specificity was also evaluated with more diverse templates from field samples.

Materials and methods

Primer design

All fungal ITS sequences at NCBI-INSO were downloaded, and the 5.8S region was extracted and aligned using the BLASTN algorithm with a gap open penalty of 25 and an ascomycete query sequence, resulting in 138 546 sequences aligned across 146 bp (Fig. 1). The alignment was compared with 5.8S sequences from selected plants, and two regions were chosen as targets for new primers (Fig. 1, Tables 1 and 2). One of the new primers, which we designate fITS9 (f stands for 'fungi'), overlaps partly in target site with the ITS3 primer (White *et al.*, 1990; Fig. 1). The fITS9 primer is degenerated at two positions and contains two inosine residues (Table 1). For the other target site, two different primer versions were constructed. The gITS7 primer is degenerated at two positions (g stands either for 'general' or for the degeneration to include G at position 13), whereas the fITS7 primer is degenerated at one position only, to obtain higher specificity for fungi (Table 2). Primers were tested on various fungal templates, and optimal primer concentrations and annealing temperatures were established by real-time PCR using the SYBR Green PCR kit (Life Technologies, Carlsbad, CA) on an iQ5 system (Bio-Rad, Hercules, CA).

Artificial template

Eleven fungal species, well distributed over the fungal phylogeny, were selected. To test for the effects of primer mismatches on amplification, some species known from sequence databases to have mismatches in the fITS9 primer site were included (Table 3). DNA was extracted from fungal cultures or sporocarps according to the following protocol: a small amount of fungal tissue was added to 1 mL extraction buffer (3% CTAB, 2.6 M NaCl, 0.15 M Tris-HCl, 2 mM EDTA, pH 8) and homogenized with a pestle. After 1 h incubation at 65 °C, spinning, extraction with chloroform and precipitation by equal volume of isopropanol, the pellet was washed with ethanol and dissolved in 50 µL of water. The entire ITS regions and flanking parts of the SSU and LSU was PCR amplified as 773–1353 bp fragments using primers according to Table 3. As these primers target sites outside the ITS1f and ITS4 primers, the long fragments included primer sites for all primers used in subsequent tests. PCR amplification was conducted in a 2720 Thermal Cycler (Life Technologies) in 50 µL reactions [0.25 ng µL⁻¹ template, 200 µM of each nucleotide, 2.75 mM MgCl₂,

primers at 200 nM, 0.025 U µL⁻¹ polymerase (DreamTaq Green, Thermo Scientific, Waltham, MA) in buffer, 5 min at 94 °C; 35 cycles of (30 s at 94 °C; 30 s at 50–67 °C; 30 s at 72 °C); 7 min at 72 °C]. The amplicons were purified using the GeneJet PCR Purification kit (Thermo Scientific, Waltham, MA), and their concentration determined with a Qubit Fluorometer (Life Technologies). The entire fragments were sequenced by Macrogen Inc., Seoul, Korea (NCBI accession numbers are provided in Table 3).

Two mixed templates were assembled by adding the PCR products from all species: (1) all species were added at equal concentrations or (2) species were added with an order of magnitude difference in concentration between them, so that the most abundant template in the mixture had 10⁴ times higher concentration than the least abundant. PCR was conducted using the ITS4 primer (White *et al.*, 1990) extended with 8 bp sample identification tags in combination with either ITS1f (Gardes & Bruns, 1993), gITS7 or fITS9. The sample identifying tags were designed using the BARCRAWL software (Frank, 2009).

Each of the two artificial templates was amplified with the three different forward primers and 20 technical replicates, totalling 120 PCRs, all with unique identification tags. PCR amplification was conducted as described previously but with only 25 cycles as well as different primer concentrations (300 nM tagged ITS4 and either of 300 nM ITS1f, 500 nM gITS7, or 1000 nM fITS9) and annealing temperatures (58 °C for ITS1f; 56 °C for gITS7; 55 °C for fITS9). PCR products were purified using the AMPure kit (Beckman Coulter, Brea, CA). Concentrations were established using a NanoDrop 2000 spectrophotometer (Thermo Scientific), and PCR products were mixed in equal molar proportion into a general sample, which was further purified using Agarose gel electrophoresis and the QIAquick Gel Extraction kit (Qiagen, Hilden, Germany), freeze-dried and subjected to 454-sequencing after addition of sequencing adaptors by ligation. Adaptor ligation and sequencing was performed by LGC Genomics GmbH (Berlin, Germany) on a GL FLX Titanium system (Roche, Basel, Switzerland).

Field samples

DNA was extracted from three samples each of soil, wood, wheat roots and hay. Soil samples were collected in late summer from a subarctic birch forest in northern Sweden, with each sample consisting of eight pooled cores from the organic horizon. Winter wheat roots were sampled in early spring from a field in central Sweden, with each sample consisting of 10 pooled root systems. One of the hay samples consisted of preconserved forage and the two other of postconserved haylage from a grass-dominated ley in Swe-

Table 1. Primer sequence and target site for the primer fITS9

Frequency among fungi (%)	G 99.7	A 99.8	A 99.6	C 99.9	G 99.4	C 99.2	A 99.8	G 99.8	C 99.2	G 96.9	A 99.9	A 99.9	A 97.3	T 98.3	G 99.8	C 99.0	G 99.9	A 99.9
						T 0.7			T 0.7	A 3.0			T 1.4 C 0.7 G 0.5	C 1.6		T 0.9		
fITS9 primer sequence	G	A	A	C	G	C	A	G	C	R	A	A	I	I	G	Y	G	A
<i>Pinus sylvestris</i> (AF037003)	G	A	A	C	G	T	A	C	C	G	A	A	A	T	G	C	G	A
<i>Picea glauca</i> (AF136618)	G	A	A	C	G	T	A	G	C	G	A	A	A	T	G	C	G	A
<i>Betula pendula</i> (AJ006445)	G	A	A	C	G	T	A	G	C	G	A	A	A	T	G	C	G	A
<i>Vaccinium vitis-idaea</i> (GU361898)	G	A	A	C	G	T	A	G	C	G	A	A	A	T	G	C	G	A
<i>Triticum aestivum</i> (FJ196304)	G	A	A	C	G	T	A	G	C	G	A	A	A	T	G	C	G	A
<i>Stellaria media</i> (EU785985)	G	A	A	C	G	T	A	G	C	G	A	A	A	T	G	C	G	A
<i>Dicranum fuscens</i> (HQ830331)	G	A	A	C	G	C	A	G	C	G	A	A	A	T	G	C	G	A
<i>Marchantia polymorpha</i> (AY342317)	G	A	A	C	G	C	A	G	C	G	A	A	A	T	G	C	G	A
<i>Trebouxia simplex</i> (EU558676)	G	A	A	C	G	C	A	G	C	G	A	A	A	T	G	C	G	A

Fungal nucleotide frequencies are based on close to 140 000 sequences (only frequencies > 0.5% are shown). Selected plant sequences are included for comparison.

Table 2. Primer sequences and target site for the primers gITS7 and fITS7

Frequency among fungi (%)	G 99.8	T 98.9	G 99.9	A 99.9	A 98.3	T 99.0	C 99.9	A 99.9	T 99.9	C 99.4	G 99.6	A 99.9	A 93.5	T 99.9	C 99.1	T 99.6	T 99.9	T 99.9	G 99.9
		C 1.0			G 1.6	C 0.9				T 0.5			G 6.5		T 0.8				
gITS7 primer sequence	G	T	G	A	R	T	C	A	T	C	G	A	R	T	C	T	T	T	G
fITS7 primer sequence	G	T	G	A	R	T	C	A	T	C	G	A	A	T	C	T	T	T	G
<i>Pinus sylvestris</i> (AF037003)	G	T	G	A	A	T	C	A	T	C	G	A	G	T	T	T	T	T	G
<i>Picea glauca</i> (AF136618)	G	T	G	A	A	T	C	A	T	C	G	A	G	T	T	T	T	T	G
<i>Betula pendula</i> (AJ006445)	G	T	G	A	A	C	C	A	T	C	G	A	G	T	C	T	T	T	G
<i>Vaccinium vitis-idaea</i> (GU361898)	G	T	G	A	A	C	C	A	T	C	G	A	G	T	C	T	T	T	G
<i>Triticum aestivum</i> (FJ196304)	G	C	G	A	A	C	C	A	T	C	G	A	G	T	C	T	T	T	G
<i>Stellaria media</i> (EU785985)	G	C	G	A	A	T	C	A	T	C	G	A	G	T	C	T	T	T	G
<i>Dicranum fuscens</i> (HQ830331)	G	C	G	A	A	T	C	A	T	C	G	A	G	T	C	T	T	T	G
<i>Marchantia polymorpha</i> (AY342317)	G	C	G	A	A	T	C	A	T	C	G	A	G	T	T	T	T	T	G
<i>Trebouxia simplex</i> (EU558676)	G	T	G	A	A	T	C	A	T	C	G	A	A	T	C	T	T	T	G

Fungal nucleotide frequencies are based on close to 140 000 sequences (only frequencies > 0.5% are shown). Selected plant sequences are included for comparison.

den. Wood samples were collected from decaying Norway spruce logs by drilling. From the root samples, DNA was extracted using the DNeasy Plant Mini kit (Qiagen), whereas from the other samples, DNA was extracted with CTAB buffer as described earlier. Soil samples were further purified with the Wizard DNA clean-up system (Promega,

Madison, WI), and wood samples were purified using the JetQuick DNA purification kit (Genomed GmbH, Löhne, Germany).

The DNA samples were subjected to PCR amplification and sequencing as described previously, but the primer fITS7 was also included (500 nM, 57 °C annealing

Table 3. Species used to assemble an artificial community template for primer testing

Species	Phylum, order	ITS1f-4 length (bp)	Primers*	Primer target sites†	NCBI accession
<i>Amanita crocea</i> (F)	Basidiomycota, Agaricales	624	NSA3 – NLC2	ITS1f CTTGGTCATTTAGAGGAAGTAA f/gITS7 GTGAATCATCGAATCTTTG fITS9 GAAC[A]CAGCGAAATGTGA	JQ912665
<i>Cortinarius purpurascens</i> (C)	Basidiomycota, Agaricales	719	NSA3 – NLC2	ITS1f CTTGGTCATTTAGAGGAAGTAA f/gITS7 GTGAATCATCGAATCTTTG fITS9 GAACGCAGCGAAATGCG[T]	JQ912666
<i>Sistotrema brinkmannii</i> (C)	Basidiomycota, Cantharellales	671	NSA3 – NLC2	ITS1f CTTGGTCATTTAGAGGAAGTAA f/gITS7 GTGAATCATCGAATCTTTG fITS9 GAACGCAGCGAAATGCGA	JQ912675
<i>Melampsora laricis-populina</i> (S)	Basidiomycota, Pucciniales	721	BMB CR – NLC2	ITS1f CTTGGTCATTTAGAGGAAGTAA f/gITS7 GTGAATCATCGAATCTTTG fITS9 GAAC[A]CAG[T]GAAATGTGA	JQ912668
<i>Fusarium poae</i> (C)	Ascomycota, Hypocreales	575	NSA3 – NLC2	ITS1f CT[G]GGTCATTTAGAGGAAGTAA f/gITS7 GTGAATCATCGAATCTTTG fITS9 GAACGCAGCAAATGCGA	JQ912669
<i>Hymenoscyphus albidus</i> (C)	Ascomycota, Helotiales	910	NSA3 – NLC2	ITS1f CTTGGTCATTTAGAGGAAGTAA f/gITS7 GTGAATCATCGAATCTTTG fITS9 GAACGCAGCGAAATGCGA	JQ912670
<i>Rhizina undulata</i> (C)	Ascomycota, Pezizales	662	NSA3 – NLC2	ITS1f CTTGGTCATTTAGAGGAAGTAA f/gITS7 GTGAATCATCGAATCTTTG fITS9 GAACGCAGCGAAATGCGA	JQ912674
<i>Debaryomyces hansenii</i> (C)	Ascomycota, Saccharomycetales	677	NSA3 – NLC2	ITS1f CTTGGTCATTTAGAGGAAGTAA f/gITS7 GTGAATCATCGAATCTTTG fITS9 GAACGCAGCGAAATGCGA	JQ912667
<i>Archaeorhizomyces finlayi</i> (C)	Ascomycota, Archaeorhizomycetales	515	NSA3 – NLC2	ITS1f CTTGGTCATTTAGAGGAAGTAA f/gITS7 GTGAATCATCGAATCTTTG fITS9 GAACGCAGCGAAGCGCGA	JQ912673
<i>Umbelopsis sp.</i> (C)	Mucoromycotina, Mucorales	656	BMB CR – LR1	ITS1f CTTGGTC[G]TTTAGAGGAAGTAA f/gITS7 GTGAATCATCGAATCTTTG fITS9 GAACGCAGCGAAATGCGA	JQ912671
<i>Mucor hiemales</i> (C)	Mucoromycotina, Mucorales	681	BMB CR – LR1	ITS1f CT[A]GG[CT]ATTTAGAGGAAGTAA f/gITS7 GTGAATCATCGA[G]TCTTTG fITS9 GAACG[T]AGCAAAGTGCGA	JQ912672

(F) indicates that DNA was extracted from fruit body material, whereas (C) indicates that a pure culture was used and (S) indicates that spores were used. The length of PCR products obtain with the primers ITS1f/ITS4 is indicated. For construction of the template, longer PCR products were produced using the indicated primers. Primer site sequences are indicated with mismatching bases in boxes. The shaded box in the *Mucor* f/gITS7 binding site indicates a mismatch with fITS7 but not with gITS7.

*NSA3, NLC2 Martin & Rygielwicz, 2005; BMB CR Lane *et al.*, 1985; LR1 van Tuinen *et al.*, 1998.

†All species matches the ITS4 primer site, except *A. finlayi* TCCTC[G]CTTATTGATATGC.

temperature). Each of the 12 samples was, thus, amplified using four different forward primers and with five PCR replicates. However, for the field samples, all PCR replicates used the same sample identification tag. DNA was also extracted from wood colonizing bark beetles, which are suspected to act as fungal dispersal agents. Four samples from different bark beetle species were amplified using the fITS9/ITS4 primer combination with 58 °C annealing temperature. PCR products were purified using the PCR-M kit (Viogene-Biotek, New Taipei City, Taiwan), cloned into *E. coli* cells using the TOPO-TA kit (Life Technologies), re-amplified and sequenced by Macro-gene Inc., Seoul, Korea.

Sequence analysis

Sequences were analysed using the SCATA pipeline (scata.mykopat.slu.se). Sequences with an average quality score below 20 or with score below 10 at any position were discarded using the high-quality region extraction option. Sequences from the artificial templates were trimmed at the ITS4 and gITS7 primer sites (primer sequences were also removed). By trimming at the gITS7 primer site, which was conserved among all templates in the artificial communities, all sequences were cropped to equal length, regardless of which primer was used for amplification. Thus, clustering was based on 38 bp of the LSU, the entire ITS2 region (122–245 bp) and 50–55 bp

of the 5.8S unit. Sequences were then compared for similarity using BLAST as a search engine. Pairwise alignments were scored using a scoring function with 1 in penalty for mismatch, 0 for gap opening and 1 for gap extension. Homopolymers were collapsed to 3 bp before clustering. Sequences were assembled into clusters (species) by single linkage clustering with 2.5% maximum distance allowed for sequences to enter clusters. Sequences that only occurred once in the entire data set (global singletons) were excluded in further analyses.

Variance in species relative abundances between the 20 replicated PCRs was partitioned into two components; one because of random community distortion during PCR amplification and one as a result of random sampling of amplicons for 454-sequencing. The sampling of amplicons for sequencing was considered a Bernoulli trial, and the variance in species frequency was calculated as:

$$\frac{p(1-p)}{n}$$

where p is the average frequency of the species and n is the average number of reads among the 20 replicate samples. The residual variance was assumed to be associated with random distortion of the community during PCR amplification.

After identification and trimming of the ITS4 primer and associated sample tags from the field sample sequences, the shortest sequence derived with the gITS7 primer was 211 bp (with the exception of a small number of shorter reads assigned to a *Candida* sp.). To equalize lengths, all sequences were trimmed to 210 bp. Thus, clustering was based on 38 bp of the LSU, 108–172 bp of the ITS2 region and 0–64 bp of the 5.8S unit. Clustering was performed as described previously, but with the clustering distance set to 1.5%. Representative sequences of all clusters (OTUs) were compared with all fungal sequences in the NCBI nr database by BLASTN using the MASSBLASTER (part of the PLUTO software; Abarenkov *et al.*, 2010). OTUs with no or few good matches to fungal ITS sequences were compared with the entire NCBI nr database manually, and those of nonfungal origin were removed before further analyses. To investigate differences in OTU composition between amplicons derived from the same sample but with different primers, Bray–Curtis dissimilarities were calculated in PC-ORD (v. 5.33d) based on relative abundances of amplicons. Variation in Bray–Curtis dissimilarity between different pairwise combinations of forward primers was analysed for statistical significance by ANOVA. To investigate the effect of primer choice on amplicon diversity, amplicon communities from each sample were randomly rarefied to the size of the smallest (which was always that obtained

with the ITS1f primer) using Excel 2010 (Microsoft, Redmond, WA). OTU richness, Shannon index and evenness (Shannon index/ $\ln$ OTU richness) were calculated as averages of 10 random rarefactions. Effects of forward primer on diversity parameters were tested for statistical significance by ANOVA with ‘sample’ included as an explaining variable. The analysis was repeated without ITS1f samples, to gain precision by enabling a lower degree of rarefaction. A more detailed analysis also included the interaction term primer $\times$ substrate.

Results

Artificial templates

Sequencing of the artificial communities after amplification with the primers ITS1f, fITS9 or gITS7 in combination with ITS4 yielded 670 260 reads, of which 65% passed the quality filtering. Sequence data is available as a single, nonfiltered file at the NCBI Sequence Read Archive with accession number SRA052087. A list of sample identifying tags is included as Table S1, Supporting Information. A small proportion of the reads was attributed to contamination by nontarget fungi (3.3%) and PCR chimeras (10 reads) and was removed prior to further analyses. The reads were unevenly distributed over the sequenced samples. For samples with mixed species, the number of reads per sample ranged from 80 to 14 792 (average 3471). It is noteworthy that samples labelled with identifier tags starting with A had, on average, three times as many reads as samples with tags starting with T.

Although supplied to the species mix in equal proportions, fungal species with mismatches in their primer sites all had low representation among the reads. The exception was *Fusarium poae*, which amplified well with the ITS1f primer in spite of a mismatch at the very 5' end of the primer (Table 3). *Archaeorhizomyces finlayi* was disfavoured with all primers owing to a mismatch at the ITS4 primer target site (Rosling *et al.*, 2011). When only species with perfect primer match were considered, the gITS7 and fITS9 primers amplified the different templates more evenly with a twofold difference in read abundance between the most and least abundant species. The ITS1f primer, in contrast, displayed higher variation with a 16-fold difference between the most and least abundant species (Fig. 2). For the ITS1f primer, there was a significant negative relationship between the abundance of species in the amplicon pool and amplicon length (Fig. 3). After subtraction of variance ascribed to random sampling for sequencing and excluding species with primer mismatches, the average standard deviation in species frequency between PCR replicates was 5.0%, 5.2% and 9.3% for amplicons derived with fITS9, gITS7 and ITS1f,

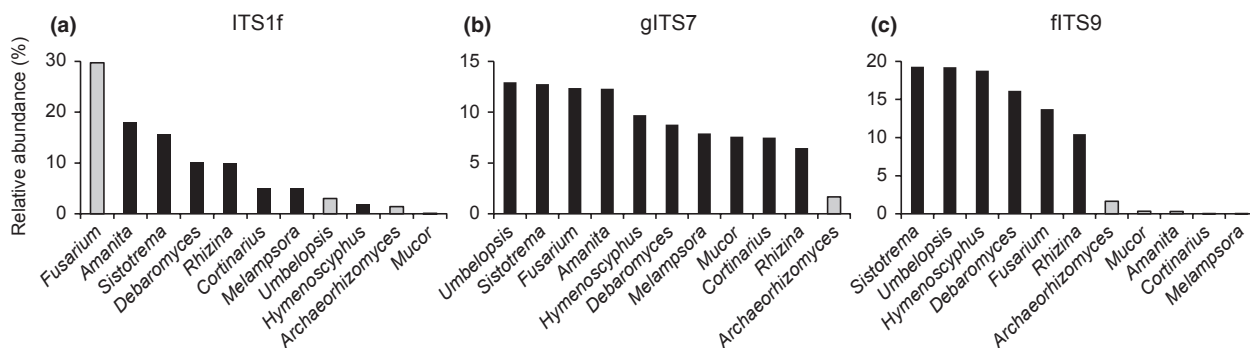


Fig. 2. Amplicon frequencies after amplification of an artificial community of fungal ITS templates with different forward primers. The different species were equally represented in the template mixture. Grey bars indicate species with mismatches in the primer binding site. (a) ITS1f – ITS4, (b) gITS7 – ITS4, (c) fITS9 – ITS4.

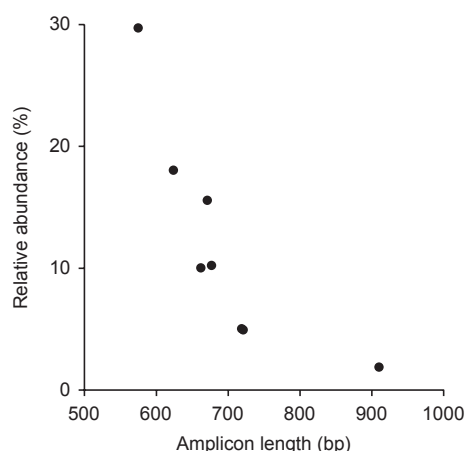


Fig. 3. Amplicon frequencies after amplification of an artificial community of fungal ITS templates with the primers ITS1f and ITS4 in relation to amplicon length. Only species with perfectly matching priming sites are included.

respectively (expressed relative to the average frequency). When the species differed in their contribution to the mixed template by four orders of magnitude, there was still a good proportionality to their relative abundance in the amplicon pool. However, the species that were added to the template mix at the lowest relative abundance (0.003%) were over-represented in the amplicon pool (Fig. 4). After subtraction of variance ascribed to random sampling for sequencing, the variation in species frequency between PCR replicates (expressed relative to the average frequency) increased with decreasing abundance of a species (Fig. 5).

Field samples

Sequencing of the field samples after amplification with the primers ITS1f, fITS9, gITS7 or fITS7 in combination with ITS4 yielded 838 739 reads, of which 73% passed

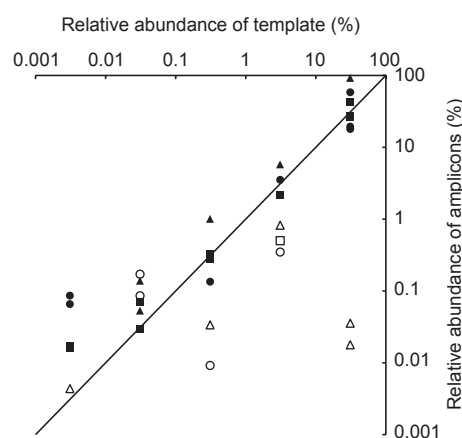


Fig. 4. Amplicon frequencies after amplification of an artificial community of fungal ITS templates with different forward primers. The different species were unequally represented in the template mixture with the most common species present at 10^4 times higher concentration than the least abundant. Circles, squares and triangles represent samples amplified with the primers ITS1f, gITS7 and fITS9, respectively. The line indicates a 1 : 1 relationship between frequency in the template and frequency in the amplicon pool. Open symbols represent species with mismatches in the primer site.

the quality filtering. Sequence data are available as de-multiplexed and quality filtered files at the NCBI Sequence Read Archive with accession number SRA052087. The proportion of nonfungal sequences was lowest in samples amplified with the ITS1f or fITS7 primers. The fITS9 primer also selected strongly for fungi. Of 174 sequenced clones with inserted PCR products obtained from bark beetles using the primers fITS9 and ITS4, all were of fungal origin (except for one, which originated from a green alga). In contrast, the gITS7 primer yielded a higher proportion of plant sequences, particularly from the hay and wheat root samples. The gITS7 primer also yielded some plant sequences from forest soil,

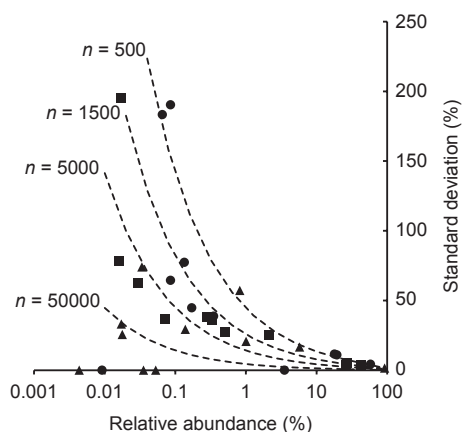


Fig. 5. Standard deviation of amplicon frequencies after amplification of an artificial community of fungal ITS templates with different forward primers. The dotted lines represent theoretical variation originating from the random sampling of amplicons for 454 sequencing (n = the total number of reads). The symbols indicate residual variation in species frequency between 20 PCR replicates after the subtraction of variance due to random sampling for sequencing. The variation is expressed as % standard deviation in relation to the average frequency of the species. Circles, squares and triangles represent samples amplified with the primers ITS1f, gITS7 and fITS9, respectively.

which were primarily attributed ericaceous species (Fig. 6).

Bray–Curtis dissimilarities were significantly different between pairwise comparisons of forward primers ($P < 0.0001$), with ITS1f amplicons diverging from the others in community composition, and fITS9, gITS7 and fITS7 amplicons being more similar (Table 4). Tested across all primers in a highly rarefied material, diversity was significantly affected by the choice of primer with respect to richness ($P < 0.0001$) but not evenness or Shannon index. A *post hoc* Tukey's test showed that the fITS7 and gITS7 primers returned significantly higher amplicon richness than the ITS1f primer. The fITS9 primer also returned higher average richness than the ITS1f primer, although the difference was not significant (Table 5). When the analysis was repeated without the ITS1f primer, richness ($P = 0.0002$) and Shannon index ($P = 0.005$) were both significantly affected by primer choice, with the fITS7 and gITS7 primers yielding significantly higher amplicon diversity than the fITS9 primer (Table 6). For richness, a significant primer $\times$ substrate interaction term indicated that the overall negative effect of using the ITS1f primer was not valid for the forest soil samples. The lower average richness returned by the fITS9 primer compared with the gITS7 primer was primarily attributed to the wood samples (Fig. S1).

When the longer sequences derived with the ITS1f primer were screened for the gITS7 primer site, mismatches

were found in 5.9% of 492 screened OTUs (singletons excluded). Ignoring mismatches at the two positions closest to the 5' end, which seem to have a minor impact on amplification (c.f. *Fusarium* in Fig. 2a), the share was reduced to 4.5%. *Saccharomycetales* was overrepresented among the OTUs with mismatches to the gITS7 primer, and a blast search showed that mismatches occurred in 30% of NCBI's *Saccharomycetales* 5.8S sequences.

When compared with ITS1f sequences, the fITS7 primer mismatched with 7.9% of the OTUs. The increase in relation to the gITS7 primer was accounted for by ascomycetes within the *Pezizomycotina*; primarily OTUs assigned to *Penicillium* species and unidentified *Orbiliiales*. The share of all NCBI sequences with mismatch to the fITS7 primer was 84% and 82% for *Penicillium* spp. and *Orbiliiales*, respectively. In addition, a screening of NCBI sequences indicated that the fITS7 primer also mismatched against 92% of sequences within the Mucorales.

When compared with ITS1f sequences, mismatches in the fITS9 primer were found in 4.1% of the 487 screened OTUs. A large proportion of the OTUs with mismatch to the fITS9 primer could not be assigned to any known fungal group because of bad matches with database sequences. Nevertheless, these OTUs had higher affinity to fungal sequences than to other organisms, and all of them also had mismatches to the gITS7 primer.

Furthermore, all three primers mismatches with members of the genera *Cantharellus* and *Tulasnella*, which are known to have rapidly evolving rDNA sequences (Moncalvo *et al.*, 2006). However, the mismatch between gITS7 and *Cantharellus* was at position 2 from the 5' end and should have little impact on amplification.

Discussion

We present three new primers, which may be used together with the ITS4 primer to amplify the ITS2 region of fungal templates. The relatively short amplicons lead to high amplification efficiency, reducing the number of required PCR cycles compared with the commonly used ITS1f primer (data not shown) and thereby minimizing distortion of community composition during PCR (Polz & Cavanaugh, 1998; Kanagawa, 2003).

Amplification and sequencing of an artificially assembled community of fungal ITS templates highlighted length variation among the PCR products as an important source of bias, which may distort quantitative relations between genotypes in a sample severely. When the entire ITS region was amplified with the commonly used ITS1f/ITS4 primer combination, there was discrimination against longer PCR products (Fig. 3), implying that the amplification efficiency of a particular genotype depends on the relative length composition of other amplicons in the sample. Quantita-

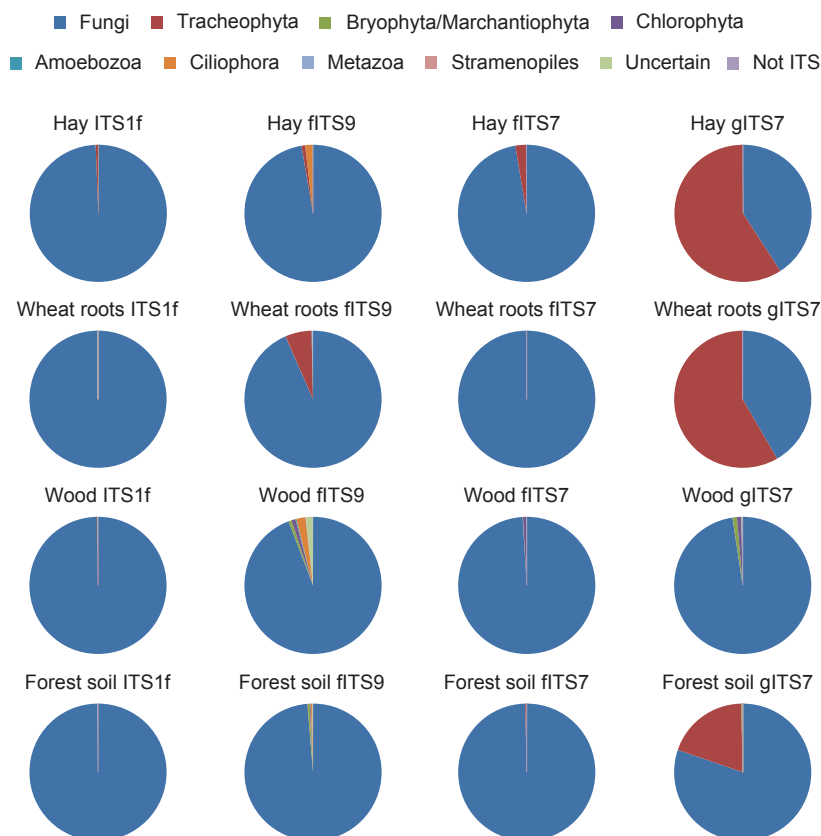


Fig. 6. Relative composition of amplicons with respect to different groups of organisms. The amplicons were derived from field substrates using different forward primers in combination with the ITS4 primer.

Table 4. Bray–Curtis dissimilarities between amplicon communities derived from specific field substrates using different forward primers

Combination of forward primers	Bray–Curtis dissimilarity
ITS1f vs. fITS9	0.32 ^a
ITS1f vs. gITS7	0.30 ^a
ITS1f vs. fITS7	0.31 ^a
fITS9 vs. gITS7	0.14 ^b
fITS9 vs. fITS7	0.13 ^b
fITS7 vs. gITS7	0.11 ^b

Figures represent average distances for twelve samples, and different letters indicate statistical significant differences between pairwise comparisons of primers according to a *post hoc* Tukey's test.

tive comparisons between samples are then capricious even for single genotypes. Thus, amplification of the entire ITS region is not to recommend, even for semi-quantitative purposes. Amend *et al.* (2010) amplified and sequenced an artificial fungal community using the primers ITS1f and ITS4 and could not find any relationship between amplicon length and frequency. This negative result, however, has to be considered in relation to a low variation in amplicon length between the species in the template mix, with five of six species having amplicon lengths within a

Table 5. Diversity parameters for amplicon communities derived from field substrates using different forward primers

Forward primer	Richness	Evenness	Shannon index
ITS1f	−7.7 ^a	0.00	−0.06
fITS9	−1.7 ^{ab}	−0.01	−0.05
fITS7	3.4 ^{bc}	0.01	0.05
gITS7	6.0 ^c	0.01	0.06

Estimates are based on random rarefaction to the size of the smallest amplicon community for each sample. Figures represent the average deviation from the over-all mean (Richness = 115, Evenness = 0.63, Shannon index = 2.95). Different letters indicate statistical significance between primers according to a *post-hoc* Tukey's test.

703–759 bp interval compared with 515–910 bp in this study.

Our results show that the length-bias problem may be overcome by using 5.8S region primers that yield shorter amplicons. When the amplified region was restricted to ITS2 only, using the newly developed fITS9 and gITS7 primers, quantitative relations between template genotypes were relatively well preserved in the amplicon community (Fig. 2b and c) even for rare community members (Fig. 4). The new primers may, thus, be used

Table 6. Diversity parameters for amplicon communities derived from field substrates using different forward primers with the ITS1f primer excluded from the analysis

Forward primer	Richness	Evenness	Shannon index
fITS9	–16.5 ^a	–0.01	–0.08 ^a
fITS7	4.1 ^b	0.00	0.03 ^b
gITS7	12.4 ^b	0.00	0.05 ^b

Estimates are based on random rarefaction to the size of the smallest amplicon community for each sample. Figures represent the average deviation from the over-all mean (Richness = 227, Evenness = 0.57, Shannon index = 3.04). Different letters indicate statistical significance between primers according to a *post hoc* Tukey's test.

for semi-quantitative analyses of between-sample differences in fungal community composition. It should be stressed, however, that translation from ITS template abundance to fungal biomass or activity is not trivial because of interspecific differences in genome ITS copy numbers and density of nuclei in biomass. Therefore, one has to be very careful when conclusions are drawn based on within-sample differences in ITS abundance between community components (Amend *et al.*, 2010).

Most likely, the length biases originated during the first PCR amplification but may have been further augmented during the emulsion PCR, which is conducted during preparation of samples for 454-sequencing. The emulsion PCR is claimed to be less prone to length biases than ordinary PCR, but when amplicons of mixed origin are subjected to 454-sequencing, it is recommended to keep the variation in amplicon length within a 150 bp span (454 Life Sciences Corp., 2011). This is possible when amplifying the ITS2 region, but not always when the entire ITS region is targeted.

The PCR cycling programs used in this test were based on standard parameter settings commonly used in many laboratories. It might be possible to overcome some of the biases experienced in connection with long PCR products by increasing the elongation time. Elongation times were not optimized in the present study, but it seems unlikely that the observed 16-fold difference in amplicon abundance ascribed to template length variation could be fully ameliorated by extending the duration of PCR cycles.

Single mismatches close to the 5' end of the primers, even 14 bp from the 3' end in the case of the *Amanita* template, drastically reduced amplification efficiency and resulted in a two orders of magnitude decrease in amplicon abundance (Fig. 2). Unexpectedly large effects of mismatches far from the 3' end have previously been observed during amplification of SSU markers from bacterial communities (Engelbrektson *et al.*, 2010). Although a primer with such a minute mismatch would be likely to

yield a fine PCR product from a single-species template, competition between templates for primers may increase the demand for primer accuracy in mixed templates. Thus, degeneration at a single position (Table 2) allowed the gITS7 primer to readily amplify various plant templates, in contrast to the fITS7 primer, which specifically amplified fungal templates (Fig. 6). The fITS9 primer selected strongly in favour of fungi because of a single mismatch with most plant templates closer to the 5' end (Table 1), but some templates from plants and protozoa were still amplified. Using the fITS9 primer, it was possible to derive nonchimeric ITS sequences from the fungal community associated with bark beetles – something which has previously been a major challenge (Y. Strid, unpublished).

The sensitivity to primer mismatches further emphasizes the risk of selection biases (Bellemain *et al.*, 2010) and makes the design of primers to amplify all members of the fungal kingdom, but yet to discriminate against nonfungal templates, a major challenge. The gITS7 primer mismatches with some species within *Saccharomycetales*, and the fITS7 primer excludes most *Penicillium* species as well as species within the *Orbiliales* and *Mucorales*. In addition, some groups of unidentifiable sequences, seemingly of fungal origin, did not match with any of the new primers. Mismatches at both of these conserved primer sites could indicate rapid evolution of the 5.8 region in some groups, similar to the situation in the genus *Tulasnella* (Moncalvo *et al.*, 2006), which is also not targeted by the new primers.

Overall, in spite of this observed discrimination against certain species or groups, *all of the new primers were equal or superior to the ITS1f primer in preserving the richness of the field templates* (Table 5). In this respect, the gITS7 and fITS7 primers outperformed the fITS9 primer (Table 6). A lower selectivity of the gITS7 primer is in agreement with its lower specificity towards fungi. The fact that the two independent primers fITS9 and gITS7 yielded amplicon communities that were more similar to each other than to the ITS1f communities (Table 4) further supports that the new primers better conserve the original composition of the template. As we did not sequence the ITS1f primer binding site in the natural templates, we cannot determine to what extent the lower generality of the ITS1f primer was because of mismatches (Bellemain *et al.*, 2010) or discrimination against genotypes with long amplicons (Fig. 3), but a combination of the two factors is likely. The primers were not evaluated against members of the *Glomeromycota*, but in a BLASTN search the gITS7 primer matched 74% of glomeromycete 5.8S sequences at NCBI, whereas the corresponding figure for the fITS7 primer was 68% and only 20% for the fITS9 primer.

The higher generality of the new primers comes with a price – they contain degenerate bases. The use of degenerate primers calls for extra careful optimization of the PCR protocol, in that the number of cycles has to be minimized for each sample so that the PCR is interrupted while still in the exponential phase (Polz & Cavanaugh, 1998). Otherwise, when one of the primer variants is depleted, templates matching other variants will continue to amplify and increase their relative abundance in the product. Therefore, when using degenerate primers, one should aim for ‘weak to medium-strong’ PCR products rather than strong bands on the gel.

Even when identical templates are analysed, the communities after PCR amplification and 454-sequencing will differ because of random errors in the analyses. When expressed as a fraction of the average frequency, the variation between PCR replicates is expected to be higher for rare species because of stochastic variation in the random selection of amplicons for sequencing, and even more so when the total number of sequences from each sample is low. After correction for this sampling error, some variation remained, which has to be ascribed random distortion of community composition during PCR amplification. This error was also larger, in relation to the frequency, for rare species. For species with a relative abundance above 0.1%, the standard deviation ascribed to random PCR distortion was usually below 50%, regardless of which primer combination was used. By pooling three PCRs for each sample, the standard deviation could be reduced below 30%, which seems reasonable for most ecological studies. For species with a relative abundance above 1%, the standard deviation ascribed to random PCR distortion was lower than 25%, and results from sequencing of a single PCR would be fairly reliable. At a sequencing depth of 5000 reads per sample, the error ascribed to random sampling of amplicons is of the same magnitude as the error ascribed to random PCR distortion, and it would not seem sensible to sequence much deeper for community studies, unless a large number of PCRs are pooled from each sample to even out distortion. If the focus is on dominant community members, 1500 reads per sample would suffice.

An interesting observation is the correlation between the number of obtained 454-reads and the identity of the last base in the sample identifier tag. As it seems, the ligation of sequencing adaptors, which are adjoined to the sample identifier tag, is biased towards certain terminal sequences (Alon *et al.*, 2011) and disfavours sequences that begin with T residues. Such ligation biases do not shift the relative abundance of taxa within samples but lead to unequal representation of samples among the total reads. Unless all used tags begin with the same nucleo-

tide, this issue has to be taken into account during pooling of PCR products.

Conclusions

During PCR amplification of full-length fungal ITS fragments, community composition may be severely distorted because of discrimination against taxa with long amplicons. Using primers in the 5.8S region to amplify the ITS2 region only, such distortion may be reduced. Quantitative relations between different templates are then reasonably well preserved both for common and rare taxa, even though the relative precision is lower for less frequent taxa. Furthermore, the new primers fITS7 and gITS7 return a more diverse amplicon community than the ITS1f primer, presumably due to a combination of better nucleotide matching to primer sites and reduced discrimination against long templates. The primers fITS9 and fITS7 were found to be more or less specific to fungi, whereas the gITS7 primer also amplified many plants (but not conifers). On the other hand, the gITS7 primer yielded the most diverse amplicon communities.

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References

- Abarenkov K, Tedersoo L, Nilsson RH *et al.* (2010) PlutoF-a web based workbench for ecological and taxonomic research, with an online implementation for fungal ITS sequences. *Evol Bioinform* **6**: 189–196.
- Alon S, Vigneault F, Eminaga S, Christodoulou DC, Seidman JG, Church GM & Eisenberg E (2011) Barcoding bias in high-throughput multiplex sequencing of miRNA. *Genome Res* **21**: 1506–1511.
- Amend AS, Seifert KA & Bruns TD (2010) Quantifying microbial communities with 454 pyrosequencing: does read abundance count? *Mol Ecol* **19**: 5555–5565.
- Begerow D, Nilsson H, Unterseher M & Maier W (2010) Current state and perspectives of fungal DNA barcoding and rapid identification procedures. *Appl Microbiol Biotechnol* **87**: 99–108.
- Bellemain E, Carlsen T, Brochmann C, Coissac E, Taberlet P & Kauserud H (2010) ITS as an environmental DNA barcode

- for fungi: an *in silico* approach reveals potential PCR biases. *BMC Microbiol* **10**: 189.
- Bio-Rad Laboratories Inc (2006) *Real-Time PCR Applications Guide*. Bio-Rad Laboratories Inc., Hercules, CA.
- Engelbrektson A, Kunin V, Wrighton KC, Zvenigorodsky N, Chen F, Ochman H & Hugenholtz P (2010) Experimental factors affecting PCR-based estimates of microbial species richness and evenness. *ISME J* **4**: 642–647.
- Frank DN (2009) BARCRAWL and BARTAB: software for design and implementation of barcoded primers for highly multiplexed DNA sequencing. *BMC Bioinformatics* **10**: 362.
- Gardes M & Bruns TD (1993) ITS primers with enhanced specificity for basidiomycetes – application to the identification of mycorrhizae and rusts. *Mol Ecol* **2**: 113–118.
- Hibbett DS, Ohman A & Kirk PM (2009) Fungal ecology catches fire. *New Phytol* **184**: 279–282.
- Hibbett DS, Ohman A, Glotzer D, Nuhn M, Kirk P & Nilsson RH (2011) Progress in molecular and morphological taxon discovery in Fungi and options for formal classification of environmental sequences. *Fungal Biol Rev* **25**: 38–47.
- Horton TR & Bruns TD (2001) The molecular revolution in ectomycorrhizal ecology: peeking into the black-box. *Mol Ecol* **10**: 1855–1871.
- Huber JA, Morrison HG, Huse SM, Neal PR, Sogin ML & Welch DBM (2009) Effect of PCR amplicon size on assessments of clone library microbial diversity and community structure. *Environ Microbiol* **11**: 1292–1302.
- Kanagawa T (2003) Bias and artifacts in multitemplate polymerase chain reactions (PCR). *J Biosci Bioeng* **96**: 317–323.
- Lane DJ, Pace B, Olsen GJ, Stahl DA, Sogin ML & Pace NR (1985) Rapid-determination of 16S ribosomal-RNA sequences for phylogenetic analyses. *P Natl Acad Sci USA* **82**: 6955–6959.
- 454 Life Sciences Corp. (2011) *454 Sequencing System – Guidelines for Amplicon Experimental Design*. 454 Life Sciences Corp. (Roche), Branford.
- Margulies M, Egholm M, Altman WE *et al.* (2005) Genome sequencing in microfabricated high-density picolitre reactors. *Nature* **437**: 376–380.
- Martin KJ & Rygielwicz PT (2005) Fungal-specific PCR primers developed for analysis of the ITS region of environmental DNA extracts. *BMC Microbiol* **5**: 28.
- Moncalvo J-M, Nilsson RH, Koster B *et al.* (2006) The cantharelloid clade: dealing with incongruent gene trees and phylogenetic reconstruction methods. *Mycologia* **98**: 937–948.
- Nilsson RH, Abarenkov K, Veldre V, Nylander S, deWit P, Brosché S, Alfredsson JF, Ryberg M & Kristiansson E (2010) An open source chimera checker for the fungal ITS region. *Mol Ecol Resour* **10**: 1076–1081.
- Polz MF & Cavanaugh CM (1998) Bias in template-to-product ratios in multitemplate PCR. *Appl Environ Microbiol* **64**: 3724–3730.
- Quince C, Lanzen A, Davenport RJ & Turnbaugh PJ (2011) Removing noise from pyrosequenced amplicons. *BMC Bioinformatics* **12**: 38.
- Rosling A, Cox F, Cruz-Martinez K, Ihrmark K, Grelet G-A, Lindahl BD, Menkis A & James TY (2011) Archaeorhizomycetes: unearthing an ancient class of ubiquitous soil fungi. *Science* **333**: 876–879.
- Schoch CL, Seifert KA, Huhndorf S, Robert V, Spouge JL, Levesque CA, Chen W & Fungal Barcoding Consortium (2012) Nuclear ribosomal internal transcribed spacer (ITS) region as a universal DNA barcode marker for *Fungi*. *P Natl Acad Sci USA* **109**: 6241–6246.
- Urich T, Lanzen A, Qi J, Huson DH, Schleper C & Schuster SC (2008) Simultaneous assessment of soil microbial community structure and function through analysis of the meta-transcriptome. *PLoS ONE* **3**: e2527.
- van Tuinen D, Jacquot E, Zhao B & Gianinazzi-Pearson V (1998) Characterization of root colonization profiles by a microcosm community of arbuscular mycorrhizal fungi using 25S rDNA-targeted nested PCR. *Mol Ecol* **7**: 879–887.
- White TJ, Bruns T, Lee S & Taylor J (1990) Amplification and direct sequencing of fungal ribosomal RNA genes for phylogenetics. *PCR Protocols: A Guide to Methods and Applications* (Innis MA, Gelfand DH, Sninsky JJ & White TJ, eds), pp. 315–322. Academic Press, San Diego, CA.

Supporting Information

Additional Supporting Information may be found in the online version of this article:

Fig. S1. Interaction effects of primer choice and substrate type on OTU richness of ITS-amplicon communities. (a) With the ITS1f primer included, (b) without the ITS1f primer.

Table S1. A list of sample identifying tags.

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